

AI at the Edge: Technology Accelerated by Wi-Fi Services

Leading the Digital
Transformation with Edge AI
and Wi-Fi Networking



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Introduction

In a world driven by technological innovation today more than ever, artificial intelligence (AI) has become a ubiquitous term used almost synonymously with digital technology. AI is quickly taking its place at the core of an inevitable digital transformation that is impacting the business world as we speak. This intelligence is primarily based on how data is collected, processed, and transmitted. Over the past decade or so, we've seen massive growth in cloud computing, enabling comprehensive applicability for AI on the cloud across various industries. However, alongside all its benefits, it was just a matter of time until certain limitations in Cloud AI began to surface, including concerns about latency, reliability, bandwidth, and security. This triggered the need for another technology to power modern AI applications without being held back by such limitations. Enter Edge AI.

Edge computing is the processing of data close to its source, which enables faster data analysis and real-time decision-making, otherwise unattainable with cloud computing. Combining edge computing's efficiency and reliability with AI's capability to analyze large volumes of data in

split seconds brings about a massive potential for Edge AI in a wide array of application areas that require local, real-time processing. This has driven Edge AI to be one of the fastest-growing technologies of the 2020s. In fact, analysts at Gartner estimate that, by 2025, over half of all data analyses by deep neural networks will take place at the edge. For reference, less than 1 in 10 data analyses was at the edge in 2021 ¹. This is also reflected in the market. Multiple market research reports on Edge AI have indicated compound annual growth rate (CAGR) estimates ranging from 21% to 24% by the end of the decade, expecting the market to cross the threshold of USD 60 billion ^{2 3 4}. One report highlights the tremendous growth in IoT device usage globally - among other industries, such as autonomous vehicles and robotics - which would propel the Edge AI market to an estimated USD 107 billion by 2029 at a CAGR of 31.7% ⁵.

In today's fast-paced technological landscape, Edge AI stands at the forefront of the digital transformation we are witnessing, especially in how it enables businesses to deploy and benefit from edge applications.

“Relay2 is pioneering the convergence of Wi-Fi and Edge Computing, fortifying industries to provide smarter and better-connected services, enhanced security, safety, and privacy, along with operational efficiency and cost optimization.”

Eric Chen,
Founder & CEO at Relay2

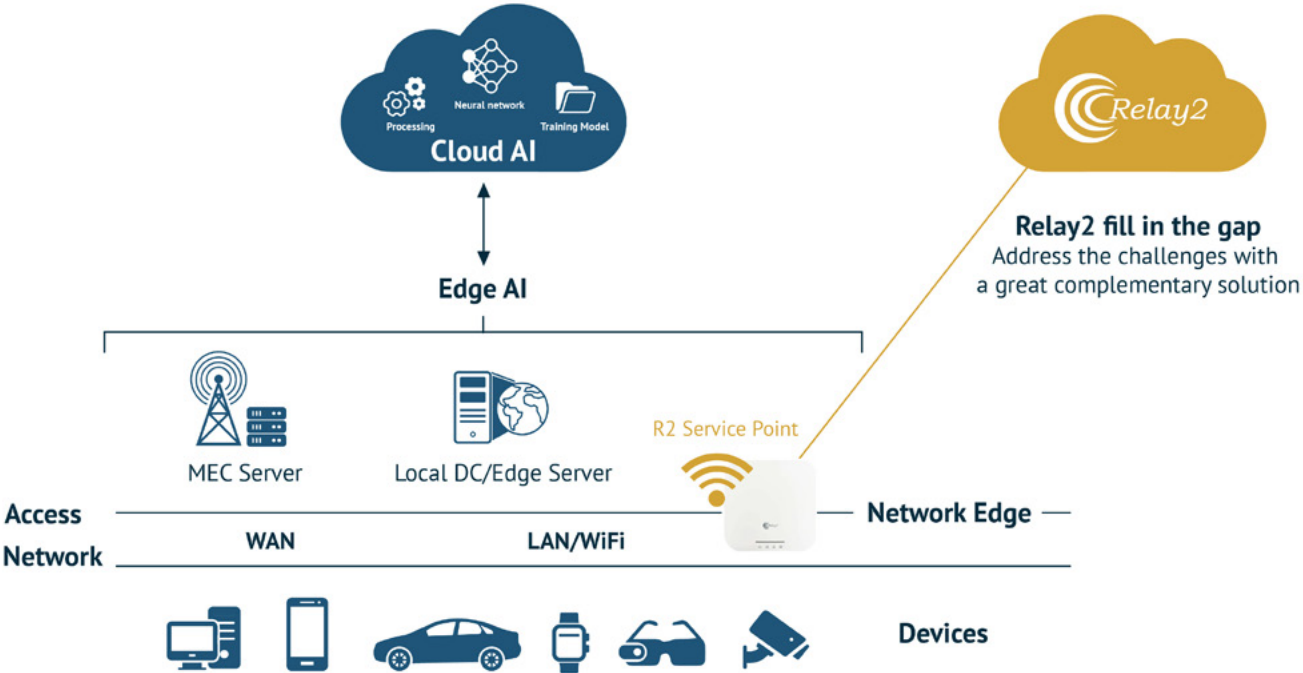
These range from smart retail to smart healthcare, education, traffic management, agriculture, and many more. Businesses in almost every industry seek to increase efficiency and safety while accelerating innovation. Such goals require growth in automation and intelligence across all processes. This is a game that Edge AI excels at. However, Edge AI is not alone. As a matter of fact, Edge AI and Cloud AI can operate together in a hybrid system, which would rely on edge computing in time-sensitive cases that require real-time response and on cloud computing for processing-intensive applications that can tolerate some latency. In this hybrid space lies a technology that is driving Edge AI even further in terms of scalability, efficiency, and performance. This technology is Wi-Fi.

To successfully deploy Edge AI applications with real-time capabilities and low latency, it is crucial to have access to high bandwidth connectivity that operates at ultra-fast speeds. This is where Wi-Fi comes into play. Deploying Wi-Fi technologies enhances Edge AI's real-time analysis and reinforces its ability to proactively respond to diverse scenarios with

minimal latency and higher bandwidth efficiency. Wi-Fi deployment is further streamlined by an advanced business model called Wi-Fi-as-a-service (WaaS). WaaS provides a managed Wi-Fi infrastructure and Wi-Fi services that enable businesses to outsource wireless network management to expert service providers while they focus on their core business activities. Merging WaaS with Edge AI brings about a dynamic technological model that integrates WaaS's flexibility and scalability with Edge AI's real-time processing and operational efficiency. This is what Relay2, a leading Wi-Fi services provider, is offering the global market today. Since its inception in 2011, Relay2 has provided over 350 customers across the US, Europe, Japan, and Asia with a platform that combines Wi-Fi analytics services and edge computing to enhance the applications' performance, security, and scalability.

In this report, we will dive deeper into the convergence of Wi-Fi services and Edge AI and explore how a technology model like Wi-Fi services can accelerate the adoption of AI at the edge. We will look into how Relay2 is serving the global market and learn about its ServiceEdge Platform and

PartnerConnect Program. We will also take a good look at some notable use cases of these technologies, including education, healthcare, and retail, and finally peek into the future of Edge AI and Wi-Fi. With this report, we aim to provide a comprehensive, in-depth analysis of how marrying Edge AI with cloud-managed Wi-Fi services will drive the growth of smart and lightweight infrastructure markets and create a smarter connected world with unified wireless access and an application computing platform.



Relay2's Service Point solution enables high connectivity and processing at the network edge for AI deployment. Credit: Relay2

Chapter I: Wi-Fi Analytics Services Enabling Edge AI

Intelligence at the Edge: Bringing AI Closer to You

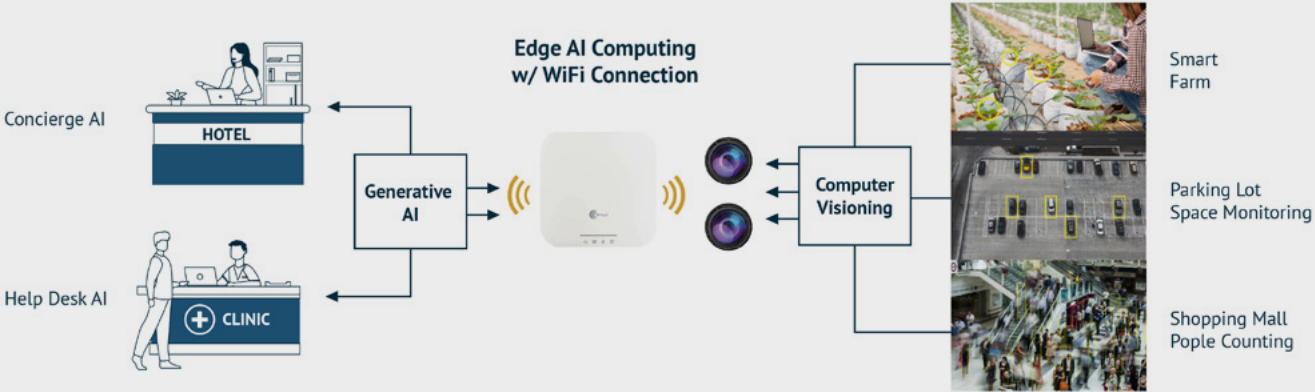
Typically, artificial intelligence has been powered by cloud computing, given AI's need for computing power and data-intensive processing. However, with the advent of edge computing, the AI landscape has been shifting, allowing for the rise of Edge AI. Edge AI is a revolutionary approach to how we deploy AI in various application areas, as it enables local computing and processing of data. That means the computing power is brought closer to the source of data, facilitating on-device data processing. With Edge AI, the system can make decisions in real time and with full contextual awareness by enabling AI features and capabilities directly on devices.

Such a process would benefit from significant advantages brought about by Edge AI, including quick response, high bandwidth efficiency, and cost-effectiveness, to name a few.

Unlike Cloud AI, which relies on remote servers for data processing and analysis, Edge AI is applied locally among a spectrum of devices spanning from data centers to IoT endpoints. The nature and location of the data being processed determine whether Cloud AI or Edge AI is more suitable for a particular application. For processing-intensive applications where latency can be compromised for the sake of in-depth data analysis and powerful computation, Cloud AI would be the ideal option. However, many applications are time-sensitive and require real-time data processing and prompt responses. This is where Edge AI takes center stage, minimizing

the need to send data back and forth to a cloud server. This can take place at any location denoted as the edge, such as retail stores, hospitals, traffic lights, autonomous vehicles, and educational facilities.

The following table explains the benefits and applicability of Edge AI and Cloud AI and how they differ from one another. Keep in mind that Edge AI and Cloud AI are not interchangeable technologies but rather applicable in different applications and scenarios. In fact, Edge AI and Cloud AI can even be complementary for systems that include both time-dependent and processing-intensive applications, which can benefit from a hybrid approach to data processing that integrates both technologies, as we'll discuss later in this report.

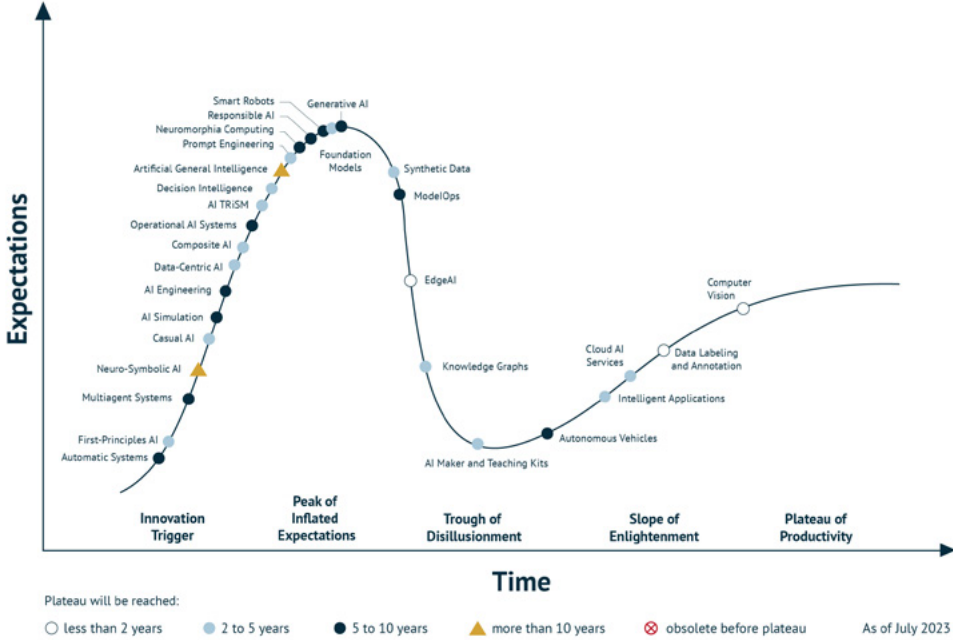


Integrating both Cloud AI and Edge AI with the help of Wi-Fi connection can enable both time-sensitive and processing-intensive applications at the same time. Credit: Relay2

Table 1: Applicability and Benefits of Edge AI and Cloud AI

Edge AI	Cloud AI
Applicability	
Time-sensitive applications , such as autonomous vehicles, real-time traffic management, medical diagnostics, and object identification and classification.	Data-intensive applications , such as big data analytics, AI-powered industrial automation, machine learning, data ecosystems, and Wi-Fi services.
Benefits	
Low latency: The amount of data transfers over wide area networks is limited with Edge AI, and thus, data processing is faster, and latency is reduced.	High computation capabilities: Cloud AI relies on large, centralized data centers that can store and compute significantly large volumes of data.
Real-time performance: The lower latency renders Edge AI suitable for applications that require real-time processing and cannot tolerate delays from data transfer.	Processing-intensive operation: Cloud AI allows for the storage and processing of large amounts of unstructured data at once, suitable for non-time-sensitive applications.
Power efficiency and reduced CO₂ footprint: Edge AI is intrinsically power-efficient and does not require intensive processing that would otherwise result in significant carbon emissions.	Flexibility and Scalability: With Cloud AI, accessibility to data from anywhere is possible, enabling flexible workflows and scalable systems without worrying about constant management of infrastructure.
Enhanced security and privacy control: With Edge AI, the amount of data exposure is minimal, thus reducing the probability of malicious security attacks and data breaches. Edge AI deployments establish a trusted data management environment that offers strong privacy control measures.	Advanced architecture and deep learning models: Cloud AI facilitates the deployment of machine learning models effectively via sophisticated architecture and pre-built algorithms that support data-intensive applications, such as Computer Vision and Natural Language Processing.
On-device learning: Some Edge AI applications are performed on a single device, enabling SoC-like devices to learn on the device without resorting to model retraining on the cloud.	Deep insights: Cloud AI enables in-depth and long-term data analyses by running through extensively large volumes of data, comparing historical data with recent data, and establishing patterns and trends.
Cost-effectiveness: Since Edge AI applications process low amounts of data, they consume less energy and can be deployed and operated at lower costs.	Cost-effectiveness: Given its applicability to data-intensive applications, the cloud offers storage infrastructure management that would otherwise be relatively costly.

Hype Cycle for Artificial Intelligence, 2023



The hype cycle for AI in 2023 showed the rapid adoption rate of Edge AI. Credit: Gartner

The Evolution and Adoption of Edge AI

A 2023 report by Gartner on innovations in and around AI has highlighted the AI hype cycle for this year, showcasing the various advancements in AI and where they are today in terms of the public's expectations ⁶. Among these innovations, we can see Cloud AI slowly navigating the so-called "Slope of Enlightenment" stage, where it is gradually transitioning toward mainstream adoption. In fact, Gartner expects Cloud AI to become mainstream in two to five years. Meanwhile, one of the fastest-growing innovations we can pinpoint on the

cycle graph is Edge AI, showing a fast adoption rate to reach mainstream productivity in less than two years. This validates the potential that Edge AI has demonstrated in the past 2-3 years and shows that both edge and cloud computing can support AI applications in a symbiotic manner in the very near future.

The recent developments in Edge AI have opened up new opportunities for devices to operate with human-like intelligence, regardless of their location. The successful deployment of AI models at the edge can be attributed to innovations that have significantly contributed to today's digital transformation. For instance, neural networks and AI infrastructure have reached a level of maturity enough to pave the way

for generalized machine learning. Organizations are now able to train AI models and deploy them in production at the edge. Besides, the recent advanced Edge AI accelerator chipset development, such as Axelera Metis, with high performance and low power budget, has made it possible to integrate these AI capabilities at a low power budget and with very high inference speed. Relay2 is pioneering this embedded integration with Wi-Fi Access Point, the edge of the Wi-Fi network. This is further strengthened by the widespread adoption of IoT devices, which is arguably the key reason behind the blowup of big data. Businesses can now collect every type of data from various sources. This availability of data and corresponding devices has made it possible to deploy AI models successfully at the edge.

“With decentralized AI processing, improved responsiveness, enhanced privacy and security, cost-efficiency, scalability, and distributed computing, Edge AI is revolutionizing our world as we speak.”

Samir Jaber,
Editor-in-Chief

Given their remarkable ability to process and comprehend a range of unstructured data, including visuals, audio, temperature, language, faces, and more, AI algorithms have emerged as effective solutions for problems that end users face in real-time. Yet, deploying these AI applications primarily in centralized data centers like the cloud may prove impractical, especially with concerns about latency, bandwidth, and privacy. This is where the benefits of edge applications can facilitate AI deployment with their localized nature and decentralization, enabling real-time insights, high availability, and enhanced privacy.

Nonetheless, Edge AI is not free of limitations, either. It still faces challenges, especially in data management, integration, security, miniaturization, and scalability. To overcome these challenges, a harmonious and purposeful co-design of hardware and software is necessary, and coordination with other technologies can prove quite fruitful. In fact, one of the most promising solutions for Edge Intelligence comes in the form of the convergence of Edge AI with cloud-based Wi-Fi Services, as we'll see in the following section. Achieving miniaturization, model accuracy, energy efficiency, and cost-effectiveness simultaneously is a complex challenge. However, with

proper coordination, optimization, and communication, we can facilitate the emergence of the next set of AI advancements at the edge.

Next-Level Intelligence: Reinforcing Edge AI with Wi-Fi Services

Since its inception about two and a half decades ago, Wi-Fi has shown significant development and adoption, with connection speeds skyrocketing from a few megabits to multi-gigabits per second. With this increase in speed, we have also seen an exponential rise in Wi-Fi-enabled devices, with almost every mobile device and computer now including Wi-Fi by default. In order to support the growing usage, demands, and throughputs of Wi-Fi, the network architecture has played a crucial role. Initially, Wi-Fi deployments were based on standalone access point architectures but later shifted to centralized hardware-based controller architectures. However, along with Wi-Fi functionalities like network optimization and mobility came some challenges, including scalability, reliability, manageability, and bottlenecks in performance, especially

with increasing Wi-Fi traffic and link speeds. To tackle these challenges, service providers like Relay2 have been harnessing the power of the cloud and enabling vendors to shift from on-premise hardware controllers to cloud-native wireless controllers. Such a cloud-native system can provide the same benefits of centralized management without the constraints of physical hardware or associated adaptations. This not only enables greater flexibility but also offers more efficient and cost-effective management and coordinated control. As a result, this space has allowed for the rise of Wi-Fi-as-a-Service (WaaS), a subscription-based service that integrates software, infrastructure, and managed services to enable centralized management and coordinated control across different access points and automation of network monitoring and troubleshooting.

Bringing Wi-Fi services and Edge AI into the same picture, we can see how their benefits align, where Wi-Fi enhances Edge AI's scalability and integration capabilities, and Edge AI furthers the user experience of Wi-Fi solutions with data-driven insights and real-time analysis. For instance, looking at Edge AI's integration challenges, we can still see cases of incompatibility stemming from differences in hardware, software, and communication protocols, which



\$69.69B (2021) - \$288.53B (2030)



\$848B (2023) - \$3.74T (2030)



0.6B Units (2023) - 1.99B Units (2028)



\$320.9b (2022) - \$1.56T (2032)

result from the unique architecture, specifications, and interface of every edge device. Employing Wi-Fi at this level enables secure, wide-coverage wireless communications, which further promotes interoperability and minimizes compatibility issues. Similarly, scalability challenges in terms of how Edge AI systems are handling increasing amounts of data and devices may compromise efficiency. The adaptability and scalability brought about by Wi-Fi services can help overcome these hurdles by enlarging a business's network infrastructure and enabling effortless infrastructure management to meet the growing demand and shifting requirements.

In other words, tying wireless and edge computing together can empower AI applications and reshape the business landscape to drive the digital

transformation we are experiencing today. This convergence of Wi-Fi and edge computing presents a promising frontier for innovation, efficiency, and growth. As companies continue to recognize the value of this convergence, we can anticipate a marked increase in its adoption. As a result, edge computing would bring about a new wave of intelligent AI applications by integrating more computing power, which would allow adaptive learning and real-time responsiveness. This would be a significant step towards a more resilient system as the demand for always-on services continues to grow across various industries and application areas, including education, smart buildings, intelligent traffic management, smart agriculture, and smart retail. As more companies embrace the convergence of Wi-Fi and edge computing, we can expect to see

a new era of technological innovation and efficiency.

Right at the center of this technological fusion is Relay2, a leading Wi-Fi service provider that is on a mission to challenge the status quo of today's Wi-Fi access point market. Relay2 is at the vanguard of the Wi-Fi edge-computing revolution, powered by their innovative ServiceEdge Platform, a Service Points technology that provides unparalleled performance, scalability, security, and privacy across a wide range of edge applications. In the next chapter, we will dive deeper into Relay2's Wi-Fi services and explore how Relay2 is poised to reshape the Wi-Fi industry and how its patented platform is enabling more AI applications and capabilities within the edge-computing framework.

Chapter II:

Relay2 and Edge AI

Wi-Fi and Edge Computing: Transforming AI Applications

The convergence of Wi-Fi and edge computing stands as a formidable force driving the ongoing shift to digital today. This convergence presents a pioneering paradigm for connectivity. It empowers enterprises with the capabilities of both managed Wi-Fi services and on-device data processing at the edge, enabling them to stay in step with, if not one step ahead of, the evolving digital landscape.

Wi-Fi analytics services, in essence, revolutionize Wi-Fi deployment through a cloud-based service model, offering a complete package of managed Wi-Fi infrastructure and services. This strategic approach spares organizations from the significant capital investments

typically associated with Wi-Fi infrastructure, affording them the opportunity to harness the expertise of service providers for broad network management. As a result, businesses can concentrate on their primary operations without having to worry about the intricacies of network maintenance. Simultaneously, the advent of edge computing ushers in a paradigm shift by migrating computational processes to the data source, thereby reducing latency and enabling real-time analytics and decision-making. This strategic switch from centralized computing models results in a notable enhancement of response time and operational efficiency, thanks to reduced bandwidth requirements and lower latency.

At the heart of this technological transformation are the edge applications, which play a pivotal role within the Wi-Fi-edge-computing space. These applications harness the potential of edge computing for localized data processing and analytics,

giving rise to real-time insights, seamless integration of artificial intelligence, advancements in IoT, and immersive user experiences. AI applications at the edge can accelerate innovation and enable organizations from various sectors to upgrade their services and maximize their operational efficiency.



Combining Wi-Fi with Edge AI provides a plethora of benefits that can be summarized in the following three points:

1. Empowering Organizational Agility: The convergence of Wi-Fi and Edge Computing offers a dynamic solution that equips organizations with the ability to scale their Wi-Fi infrastructure in response to ever-evolving demands. By executing localized data processing and applications at the edge reinforced with Wi-Fi Analytics Services, businesses can seamlessly expand their network infrastructure and minimize their reliance on centralized data processing, thus optimizing bandwidth utilization and curtailing expenses.
2. Improving Cost-Efficiency and Streamlining Infrastructure Management: With Wi-Fi Networking and Edge Computing convergence, organizations can delegate the management and

upkeep of Wi-Fi networks to service providers, hence alleviating the complexities of network infrastructure management while reaping the benefits of trustworthy and proficient Wi-Fi services. This also minimizes the initial financial burdens associated with building and maintaining Wi-Fi infrastructure. This integration yields the merits of both a cloud-managed Wi-Fi framework and local data processing, delivering immediate data insights and facilitating real-time decision-making.

3. Enhancing Performance and Reducing Latency: Leveraging local data processing at the network edge significantly mitigates latency in comparison to transmitting data across long distances to the cloud. This, in turn, results in prompt response times, enriched user experiences, and substantial improvements in application performance.

In other words, Wi-Fi Services with Edge Computing emerges as a pivotal catalyst for digital transformation. Through its localized data processing capabilities, it delivers faster (low latency), more efficient (less bandwidth usage), and highly secure (enhanced security) data handling to enable real-time decision-making, which is a central factor of many applications, including autonomous driving, healthcare, and smart factories.

Relay2 ServiceEdge Platform: Reshaping Wi-Fi Access Point Technology

In a recent study by Gartner, analysts estimated that by 2025, 75% of all enterprise-generated data will be created and processed outside of the cloud or traditional centralized data centers ⁷. These predictions highlight the significance of edge computing and its growth in enabling AI applications at the edge. This is where Relay2's Service Point technology emerges, providing a state-of-the-art, all-in-one solution that combines Wi-Fi services with edge computing.

Relay2's Service Point technology marks a substantial advancement in the field of Wi-Fi access point technology, pushing the boundaries of what conventional access points can achieve. Traditional access points may offer fundamental connectivity, yet they frequently fall short in providing the essential features required to fully exploit the potential of edge computing. The Relay2 Wi-Fi Service Point is a high-caliber Wi-Fi access point that incorporates inherent edge computing capabilities. It offers the capacity to host and execute local edge applications, which are either native software applications or equivalent counterparts of cloud-based applications. This convergence of edge computing at the Wi-Fi access point results in enhanced application and service performance, real-time

interaction and response, and secure data processing, ensuring privacy and confidentiality.

Using Relay2's Service Point, businesses can leverage Wi-Fi services with edge computing to reap the following benefits:

- 1. **Efficient use of extra computing resources:** By delegating computational duties from centralized servers to the Service Point devices, enterprises can enhance their network's overall performance and diminish latency. These devices have powerful processors and extensive memory, which enable streamlined handling of complex computing tasks at the edge of the network.

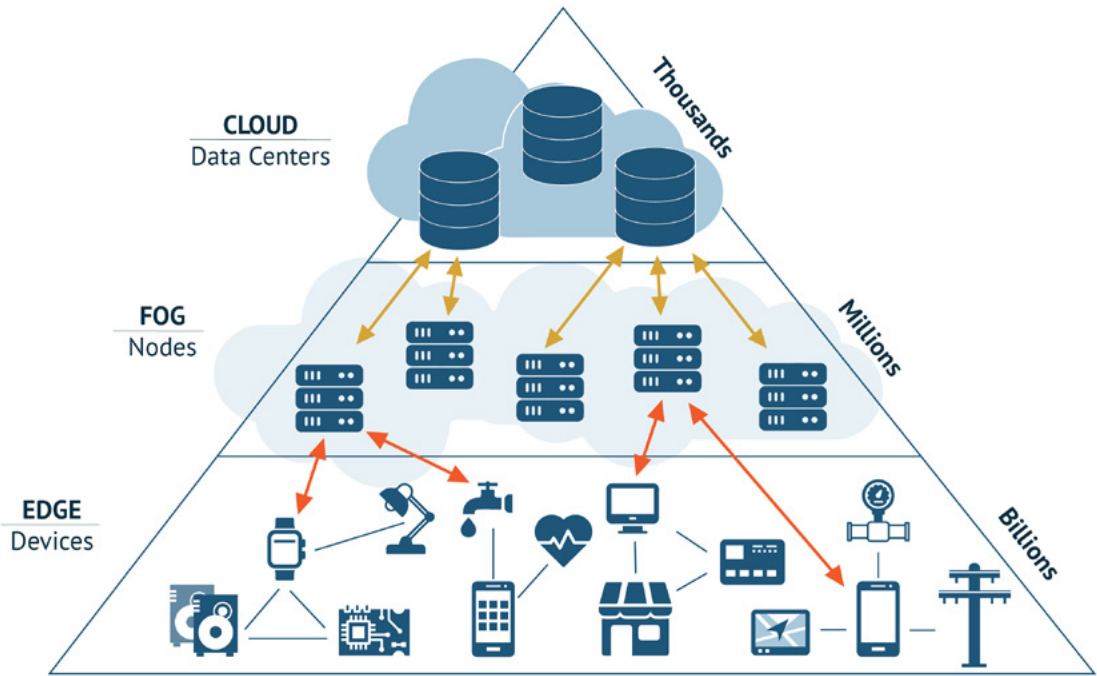
- 2. **Enhanced performance, reliability, and security:** Service Point devices have built-in security features and advanced encryption protocols that support organizations' privacy and integrity of their data. Furthermore, these devices are engineered to manage substantial data workloads and sustain unwavering performance, even in challenging operational settings.
- 3. **Streamlining infrastructure and achieving sustainable cost efficiency:** Businesses can significantly benefit from an all-in-one device that integrates switches, routers, and computer servers, enabling them to optimize their network infrastructure and minimize long-term operational and maintenance costs. This comprehensive

approach simplifies deployment and management processes, granting organizations the ability to allocate their resources with increased efficiency.

Relay2 has taken one step further and has developed a cutting-edge platform called the ServiceEdge Platform. This platform is purpose-built to coordinate and oversee Relay2 Service Points and the edge-based applications they host, all orchestrated seamlessly from the cloud. The Relay2 ServiceEdge Platform comprises a network edge device (the Relay2 Wi-Fi Service Point), combined with a cloud portal service that allows for the management and coordination of Wi-Fi networking, Service Point, and edge computing. This platform enables the distribution, hosting, provisioning, and automation

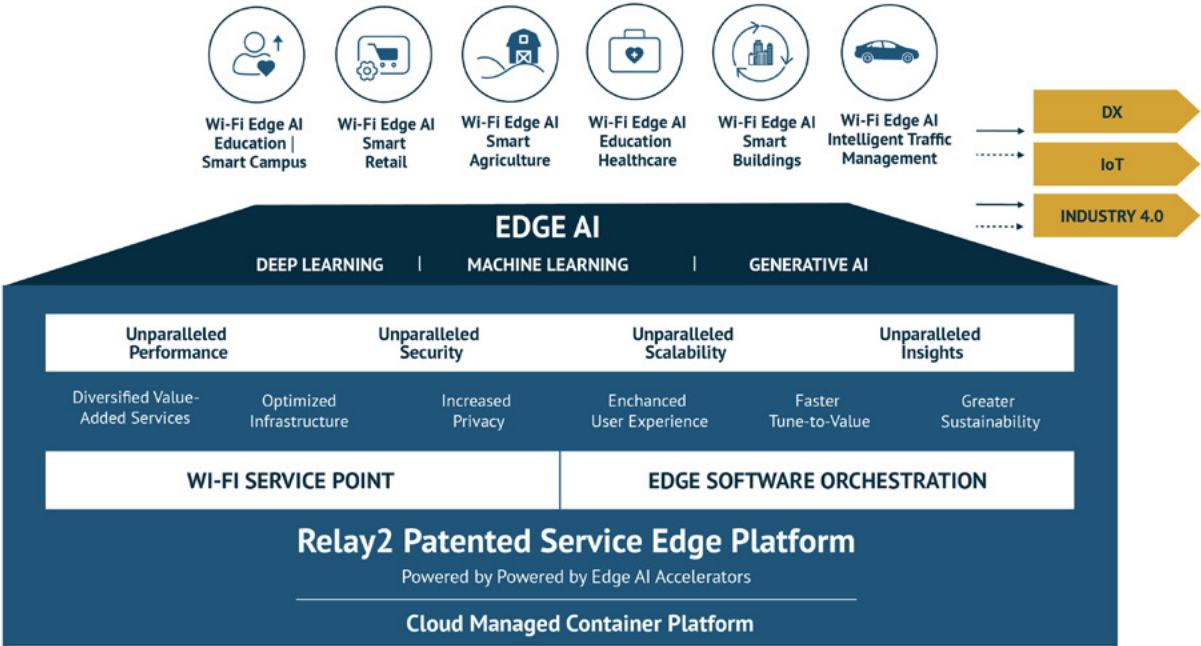
of custom-built or open-source edge applications via the Edge Apps Orchestrator. The overarching goal of the Relay2 ServiceEdge Platform is to offer a more efficient solution for WaaS providers to overcome the challenges of increasing operating costs and diminished ROI. By incorporating edge computing for advanced value-added services, this platform seeks to enhance the competitiveness of WaaS providers in the marketplace.

Going beyond the conventional Wi-Fi access points focused on network connectivity, Relay2 Service Points introduce advanced edge-computing capabilities. These capabilities enable the hosting of edge applications within the Wi-Fi network in closest proximity to users and data sources. Relay2 expands the reach of edge computing,



Representation of a distributed artificial intelligence layered architecture (Credit: Yash Agrawal).

Relay2 Patented ServiceEdge Platform



The Relay2 Patented ServiceEdge Platform showing how the Relay2 Service Points can support Edge AI applicability. Credit: Relay2

providing an opportunity for cloud computing applications to leverage the deployment of their edge-computing equivalents.

The Relay2 Edge Apps Orchestrator stands out in the market due to a set of unique and compelling features. It introduces hierarchical, tiered-account cloud management services tailored to distributors, service providers, and end customers, facilitating Cloud Wi-Fi network management for Managed Service Providers (MSPs) while reducing operational expenses. Notably, the Edge Application Store within the Relay2 Wi-Fi Cloud Manager presents a distinct advantage, allowing MSPs to swiftly download and install prequalified applications for Service Points, expediting proof-of-concept for value-added services. Furthermore, Relay2 offers a cloud portal for

provisioning, management, and automation of application hosting and settings on Service Points, enhancing user convenience through centralized configuration and administration. The platform also seamlessly orchestrates docker containers with its patented edge container system, streamlining the dynamic deployment of containerized edge applications while optimizing resource utilization and simplifying application management.

Leveraging docker containers for the development of edge applications on Service Points, Relay2 fosters a developer-friendly, open-source environment, enabling IT technicians to create and deploy applications efficiently, thereby reducing time-to-market. Lastly, the Relay2 Cloud Manager provides feature-rich graphics, user interfaces, and

monitoring capabilities, offering comprehensive insights into edge-computing applications and network performance for organizations.

With Relay2 ServiceEdge Platform, businesses and developers can take part in the digital transformation and benefit from Relay2's store of ready-made edge-computing applications to reduce their time to market and enhance their infrastructure's scalability and flexibility with unparalleled automation and orchestration. They can also boost their operational efficiency with streamlined IT operations enabled by a centralized cloud-based management portal, all the while future-proofing their digital network infrastructure and unlocking new opportunities in emerging technologies like AI and IoT.

Serving the Edge AI Market: Relay2 PartnerConnect Program

Relay2's pioneering ServiceEdge Platform offers a remarkable avenue for organizations to enhance their value proposition to customers and boost their profitability. This platform delivers a unique combination of superior application performance, robust security, scalability, and data privacy, all within a cost-effective, serverless infrastructure. Additionally, it introduces innovative revenue-generating services, positioning businesses for success in the ever-evolving landscape of Wi-Fi Services with Edge Computing.

If you are interested, Relay2 has launched a PartnerConnect Program, where they provide partners with customized services that would help distinguish their Wi-Fi with Edge Computing, strengthen their profitability channels, and deliver high value to customers. Reach out to one of Relay2's partner teams at partners@relay2.com to learn more about how you can partner with Relay2 to lead the growing, competitive Edge AI market.

Chapter III: Applications & Case Studies

The applicability of Wi-Fi Edge AI is fairly expansive and is growing by the day, spanning various industries and application areas. The most central aspects of this intersection of technologies are the real-time insights they provide, how these insights impact the speed and type of response, and how they are communicated across the network. Whether it is in smart healthcare, education, retail, or other application areas, Wi-Fi-enabled Edge AI is redefining how technology is

leveraged in real time. Relay2 utilizes several Edge AI platforms to enhance the functionality of AI applications on the edge. In this chapter, we will take a look at some interesting use cases of Wi-Fi Edge AI, and we will see the impact of this technology convergence and the new opportunities it is opening in new and emergent technological spaces.

Education / Smart Campus

Wi-Fi Edge AI plays an important role in education-related applications, among which is remote learning integration, facilitating seamless and interactive remote education and bridging the gap between on-site and remote learning experiences. With localized content engagement pattern analysis and personalized experiences for students, Wi-Fi services and edge computing can help enhance and refine education content delivery.

Campus facilities can also benefit significantly from integrating such technologies, especially in terms of security and surveillance, where Wi-Fi Edge AI can help analyze surveillance videos locally and in real time to recommend timely actions for security staff. It also powers use cases like attendance automation and traffic patterns, streamlining attendance tracking for educators and student flow analysis for operation enhancement.

Another use case of Wi-Fi and edge computing can be seen in revenue generation, in which they enable intelligent, location-based advertising through audience behavior analysis for commercial partners. By leveraging these technologies, educational institutions can help promote campus services better, boost product marketing across campus premises and beyond, and enable efficient digital transactions.

One example of applying Wi-Fi and edge computing in education is by [WENet](#), an OEM partner of Relay2 in Vietnam. In order to support the University of Transport Technology in Vietnam in developing a digital technology infrastructure platform, WENet deployed Relay2's Wi-Fi-enabled edge computing solution, making use of the advantages provided by Relay2 ServiceEdge Platform compared to conventional Wi-Fi networks. From stable bandwidth to fast processing speed, short deployment time, minimal service interruption time, and optimal

operating costs, the features brought about by Relay2's ServiceEdge Platform were highly effective. As a result, over 17 thousand students across three campuses were able to benefit from this platform, especially in terms of the following three key functions:

1. Digital signage interacting with mobile devices to support two-way communication (university-to-student communication and student responses to university requests)
2. Students' access to digital education materials from Relay2's Service Points
3. Students' ability to order food digitally from the cafeteria

Such functions have not only enabled students to benefit from a Wi-Fi-powered edge computing platform but also helped collect analytic data for future AI processing at the edge. Read more about the [WENet use case](#) on the Relay2 website.

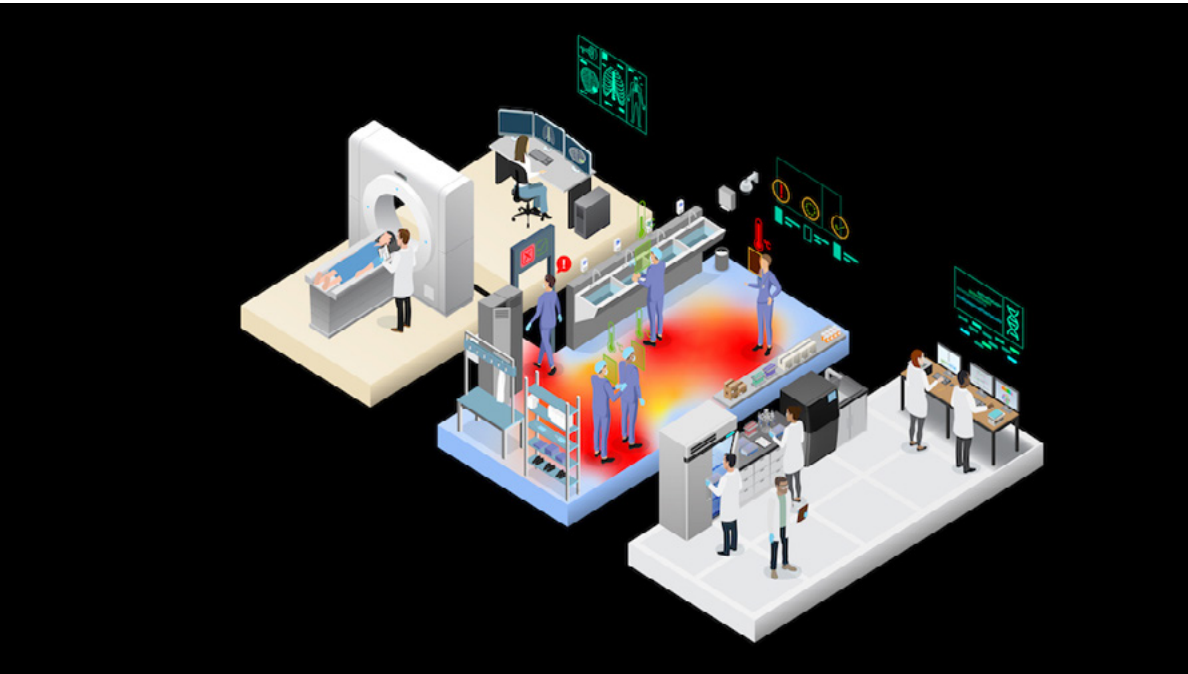


Smart Healthcare

Undoubtedly, one of the most important aspects of healthcare is real-time monitoring and real-time decision-making. However, as we have seen over the recent years with pandemic-caused crises, healthcare workers are more often than not overworked, and hospitals are resource-limited, which can limit their abilities to provide timely care. In order to ensure patient safety and health within hospital spaces and elsewhere, it is crucial to track, monitor, analyze, and act in real-time. All this can be made available by incorporating Edge AI and Wi-Fi. Whether it is remote monitoring and management, patient activity tracking, patient care, or even assistance during healthcare crises, hospitals, and medical personnel can benefit significantly from integrating Wi-Fi Edge AI into their processes.

Smart healthcare comes in the form of connected healthcare, where AI-based smart devices, healthcare management software, and digital instruments are integrated into a “connected” healthcare system using a network protocol like Wi-Fi. As such, medical personnel would have access to all the medical-related information they need to provide the patients with the necessary medical treatment in the most effective and convenient way possible. This has been made possible by utilizing Edge AI and Wi-Fi technologies to empower the Internet of Medical Things (IoMT). In other words, healthcare personnel can leverage the low-latency capabilities of Edge AI processing and the ubiquity of Wi-Fi connectivity to access real-time data from diverse medical sensors, enabling timely decision-making and interventions and fostering a proactive approach to patient care.

Consider intensive care units (ICUs), operating rooms, or other healthcare spaces, the use of Edge AI and Wi-Fi can augment the work of doctors, nurses, and clinicians by minimizing limitations, redundancies, and time allocated to activities away from direct patient care (such as documentation). As a result, medical workers can benefit from an emerging ambient intelligence ⁸, powered by edge computing, AI capabilities, and Wi-Fi. This can even be expanded to daily living spaces, which would support independent living, chronic disease management, mental health, and physical rehabilitation, particularly of elderly individuals.



Edge computing is transforming healthcare. Credit: Nvidia

Smart Retail

Global retail is almost always in flux, and both retailers and suppliers are grappling with the complexities of ever-changing consumer demand, behavior, and expectations. The continuous shifts in these dynamics create uncertainties in predicting future trends and exert significant stress on international supply chains and store operations. Navigating these challenges necessitates real-time adaptability to stay ahead in the retail industry.

Edge AI emerges as a potent solution, offering retailers the capability to swiftly and precisely forecast daily demand while automating crucial aspects of supply chain logistics, customer engagement, and operational agility. This not only addresses challenges related to in-store supply

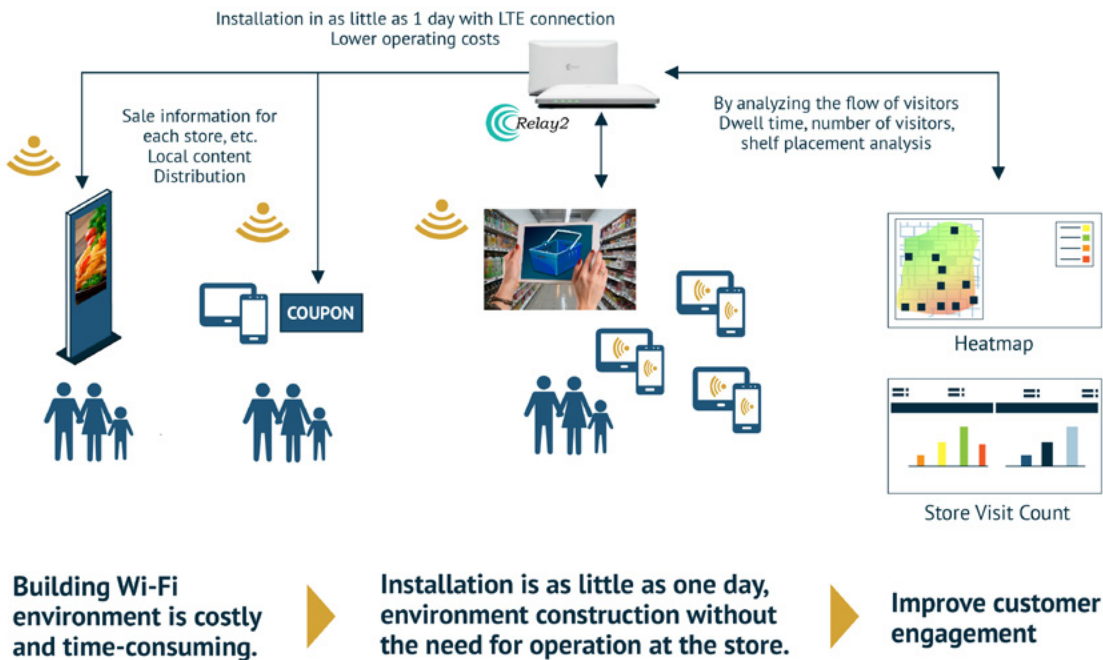
management but also tackles issues in last-mile delivery costs, aligning with consumer expectations regarding timely deliveries.

Within physical stores, retailers leverage Edge AI to combat issues such as shrinkage and stockouts, optimizing inventory management. Simultaneously, the application of Edge AI and Wi-Fi contributes to the creation of frictionless in-store experiences while prioritizing the health and safety of both employees and shoppers. The integration of AI in retail is thus transforming operations, offering a multifaceted approach to address the intricate challenges posed by the dynamic retail landscape.

D.Phone, one of the largest cell phone retailers out of China, has deployed in-store Wi-Fi to enhance its customer experience and promote engagement as a way to offer a differentiated

experience that would distinguish its D.Phone stores from competitors. They opted for Relay2’s wireless broadband solution, taking advantage of its unified native-cloud wireless networking and edge-service delivery.

By leveraging Relay2 access points, D.Phone was able to offer high-quality customer experiences and deliver value-added services in-store. With its fast and reliable coverage in demanding environments like shopping centers and large venues, Relay2 access points enabled D.Phone to meet its customers’ expectations and offer them the high-speed connectivity they desire. Moreover, with Relay2’s Edge services, D.Phone was able to run sales-generating applications and computation-intensive services locally and place content right at its D.Phone customers’ fingertips, thus reducing content



How Relay2 Service Points can be applied to enhance customer engagement in smart retail. Credit: Relay2

download time to just seconds. This helped streamline the customer purchasing experience and circumvent bandwidth bottlenecks. It even enabled D.Phone to deliver new services to its customers, including app delivery and couponing. This has displayed an accelerated go-to-market process for D.Phone’s digital merchandise and a reinforced branding strategy. Read more about the D.Phone use case on Relay2’s website.

Smart Buildings

With the recent growth in smart homes and building automation systems, edge computing, AI, and Wi-Fi have become increasingly popular in this space. The integration of Edge AI with sustainable architecture and smart buildings has been rapidly gaining momentum and transforming the manner in which we construct and interact with our built spaces. With advantages like enhanced energy efficiency, optimized systems, personalized occupant comfort, and streamlined building maintenance, smart buildings are leveraging the real-time analysis capabilities of Edge AI and its integration with IoT and developing potential for more sustainable architecture and digital integration.

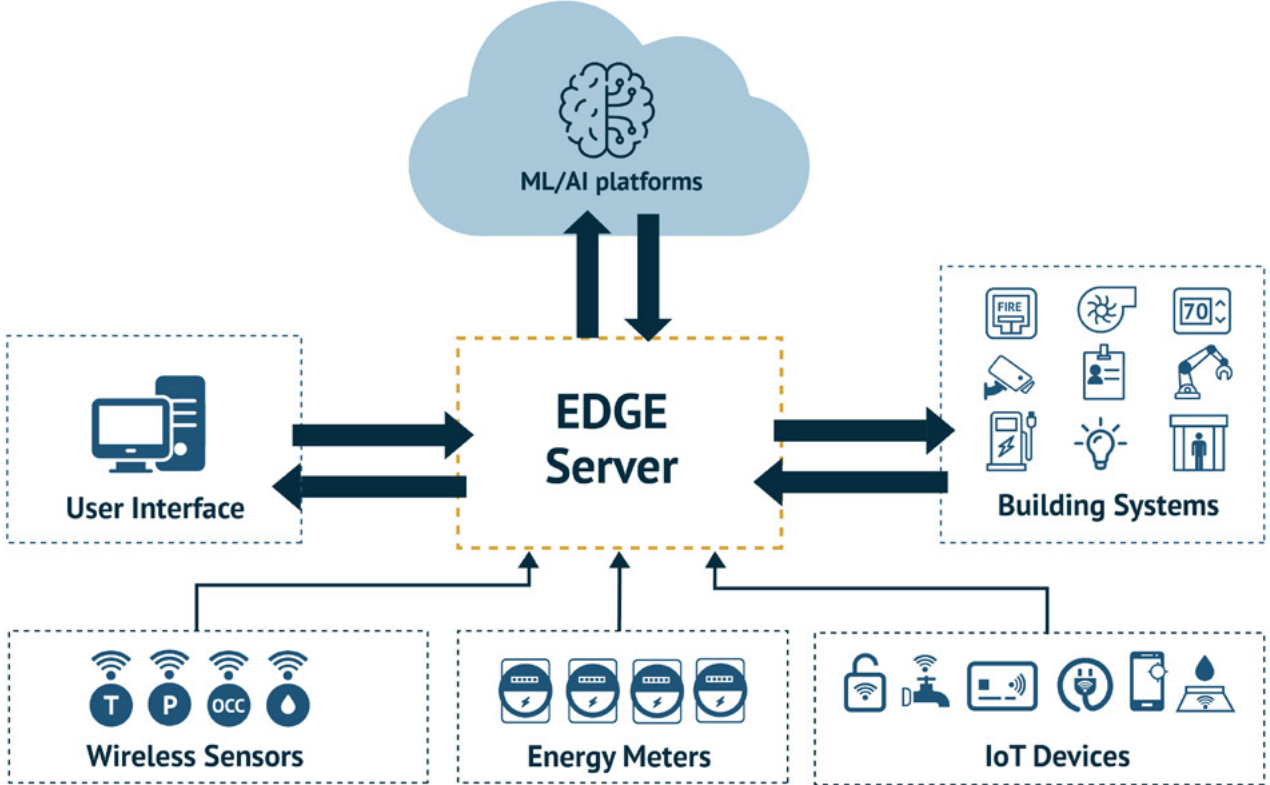
Acting as a centralized intelligence hub, AI processes and interprets data obtained from an array of edge devices (IoT devices) such as sensors and actuators scattered throughout the building. This network of devices forms a comprehensive information system, providing valuable insights into various aspects of the building’s operations. Through the implementation of AI algorithms, smart buildings gain the ability to efficiently

monitor and regulate crucial functions like lighting, heating, ventilation, air conditioning, security, and energy consumption. Such an integration results in an environment that is not only responsive but also adaptive, catering to the dynamic needs and preferences of the building occupants.

This can be further supported by Wi-Fi connectivity and cloud-based Wi-Fi services, in particular. Cloud-based Wi-Fi services enable centralized management and control of Wi-Fi networks. This centralized approach simplifies the deployment and maintenance of Wi-Fi connectivity across the building, ensuring a more streamlined and scalable infrastructure. Furthermore, the cloud facilitates the storage and analysis of the vast amounts of data generated by IoT devices in Smart Buildings. AI algorithms require substantial data inputs to make informed decisions, and cloud-based services offer the computational resources needed for processing and interpreting this data. It enables more sophisticated analytics, allowing for better insights into building operations, occupant behaviors, and energy consumption patterns.

Similarly, the cloud’s scalability can easily accommodate the dynamic nature of Smart Buildings, especially when the number of connected devices grows or changes. Just as importantly, cloud-based services enable remote monitoring, management, and updates of Wi-Fi networks and connected devices. This remote accessibility is invaluable for building administrators, allowing them to troubleshoot issues, implement updates, and optimize network performance without physically being on-site. In this

regard, smart buildings can benefit meaningfully from Relay2’s innovative solutions. With advanced networking capabilities, Wi-Fi Edge Computing power, and intelligent analytics, users of Relay2’s ServiceEdge platform can ensure seamless connectivity and optimized operational efficiency.



“Relay2’s plug-and-play access points and cloud-based platform [have] given us a solution that lowers the cost of deployment and management.”

Wu Huan,
CIO, D.Phone

Smart Agriculture

The Internet of Things (IoT) and edge computing have influenced the agricultural and farming industries quite extensively over the past decade, especially with the rapid rise of low-power wide-area networks (LPWAN) ⁹. Wireless communication protocols are, in fact, a vital part of smart agriculture systems, especially since the fields cover large areas that necessitate system components to be placed at remote locations. The integration of Wi-Fi and edge computing has been key to expanding the digital functionalities in agriculture, especially with the advent of AI at the edge, thus increasing the possibilities of smart agriculture over large areas.

One type of smart agriculture that has been propelled by Edge AI and Wi-Fi services is precision farming. By deploying sensors and cameras in fields, farmers can gather real-time data on crucial factors such as weather patterns, soil conditions, crop growth,

and pest infestations. This localized data processing enables farmers to modify and regulate their irrigation, fertilization, and pest control without delay, leading to an increase in crop yields and resource efficiency. The implementation of edge devices equipped with sensors extends beyond crop monitoring to include livestock health and behavior. Wearable devices on animals, such as cattle, provide early indications of illness or oestrus, empowering farmers to take timely actions, resulting in improved animal well-being, reduced rates of mortality, and higher farm productivity all around.

Furthermore, the incorporation of Edge AI has paved the way for autonomous farming machinery, where tractors and drones equipped with edge devices can analyze field data in real time. This enables them to refine their routes, control planting depths, and identify and eliminate weeds in an autonomous manner. Not only does autonomous farming machinery reduce

labor costs, but it also minimizes the use of chemicals, aligning agriculture with more eco-friendly practices. This is further empowered with the integration of Wi-Fi Services, allowing for centralized, remote monitoring and seamless connectivity between edge devices across large fields and in remote areas. The seamless integration of Edge AI and Wi-Fi technologies is proving transformative in making agriculture smarter, more efficient, and more sustainable over expansive agricultural landscapes.

Relay2 offers smart farms state-of-the-art solutions that can help transform crop production, efficiency, and sustainability. These solutions seamlessly connect and monitor agricultural devices, collect real-time data, and leverage advanced analytics for informed decision-making. By optimizing resources, automating processes, and implementing precision farming, Relay2 empowers farmers to maximize yield and sustainability.



Smart farming makes use of sensors at the edge for sustainable architecture. Credit: Continental

Chapter IV: Future of Edge AI with Wi-Fi Technology

The next years of this decade are going to be extremely exciting for AI, particularly Edge AI. And with Wi-Fi Services like Relay2 revolutionizing the access point market, the future is going to be very promising indeed. The surging demand for AI applications that offer real-time, low-latency, and privacy-conscious capabilities will usher in a wave of Edge AI implementations. These deployments are poised to become increasingly precise and efficient, capitalizing on advancements in communication networks, such as Wi-Fi 6, coupled with innovations in artificial intelligence (such as data-efficient AI) and the Internet of Things (IoT).

This transformative landscape underscores the pivotal role played by researchers, industry leaders, and open-source communities in steering the trajectory of Edge AI development. The evolution of efficient Edge AI applications relies on the progress of various technologies designed to support edge devices and enhance the computational efficiency of AI functions, including Wi-Fi services.

Wi-Fi 6 applications



The applications of Wi-Fi 6. Credit: Digi

Increased Adoption & Integration with Edge AI

As the demand for real-time processing and low-latency applications continues to surge, the integration of Edge AI with Wi-Fi services will follow suit. The agility and scalability of Wi-Fi seamlessly complement the requirements of edge computing and AI. For instance, the latest generation of Wi-Fi, Wi-Fi 6, promises much faster data transfer speeds that can reach up to five times the speeds of Wi-Fi 5. This is complemented by reduced power usage, increased capacity, and enhanced performance, providing

higher connection reliability in environments with multiple connected devices. This is where Relay2 sets a standard with its Wi-Fi services, pushing for innovation that leverages advancements like Wi-Fi 6. With a dynamic system that ensures flexibility and security, Relay2 offers virtual access control and simultaneous HD video sessions that can support up to 128 sessions. This is further supported by high performance and management of heavy traffic, capitalizing on an average throughput DL that exceeds 2 Gbps and handles 1,024 clients concurrently.

This is bound to grow and enhance the adoption and integration with

Edge AI in order to enrich the user experience. By implementing AI at the edge for efficient data processing and leveraging Relay2's proactive real-time analysis and natural language adaptability, businesses and organizations can elevate their user experience beyond merely improving connection reliability. This synergy allows for efficient processing closer to data sources, reducing latency and enhancing overall system responsiveness.

With Wi-Fi 7 being the next generation of Wi-Fi, we could see an even more seamless and immersive user experience and an advanced Edge AI performance, building on its new

features such as multi-link operation (MLO), 320 MHz superwide channels, and a 4k quadrature amplitude modulation (QAM). This convergence of state-of-the-art communications and information technologies, such as cloud-native Wi-Fi services, edge computing, IoT, and AI, is driving a massive leap in management efficiency and social productivity. The hybrid edge-cloud architectures and applications are likely to surge in the very near future, ensuring not only high speed and efficiency but also the ability to run multi-modal generative AI and language models at the edge ¹⁰. In the second half of this decade, we will arguably witness an unprecedented increase in the adoption of Edge AI across numerous applications, rendering the technology mainstream.

Computer Vision & the Impact of Edge AI and Wi-Fi

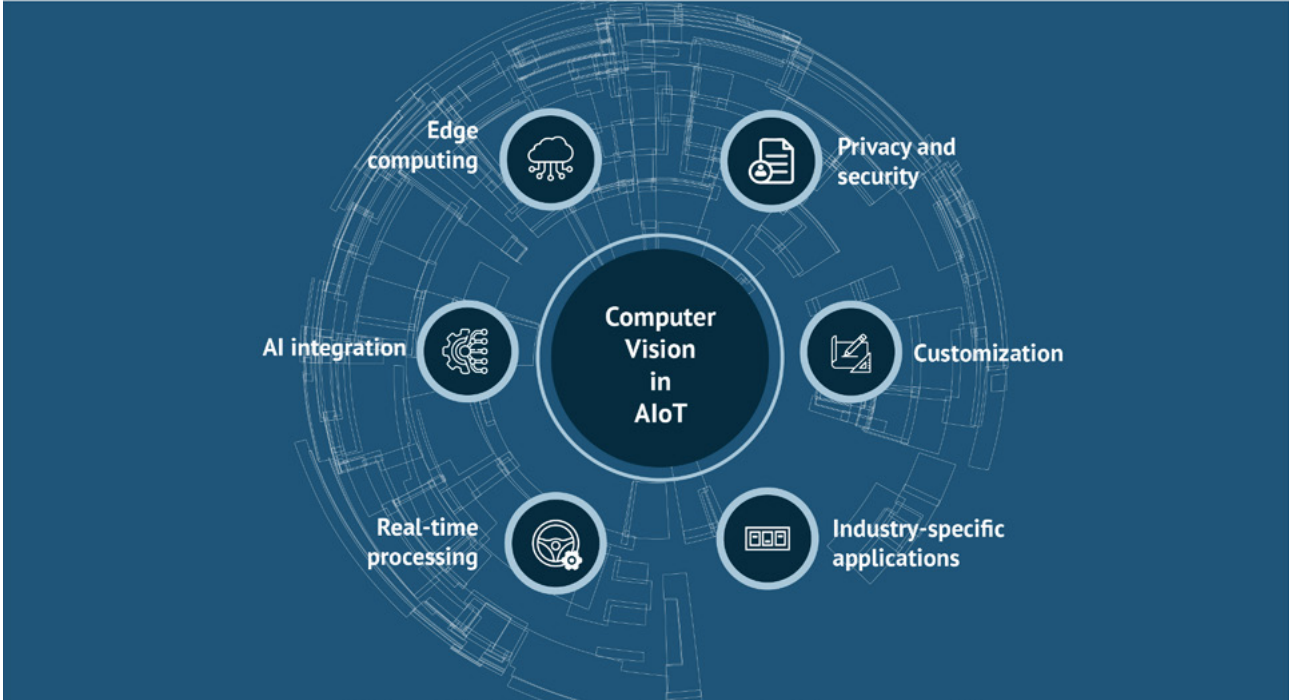
One area where the convergence of Edge AI and Wi-Fi services is poised to have a high impact is computer vision applications. With the capability to process data locally, Wi-Fi-empowered edge devices can execute intricate computer vision tasks with unparalleled speed and efficiency. This transformative shift not only minimizes reliance on centralized cloud servers but also allows positioning systems to optimize energy and costs, enabling applications such as facial recognition, object detection, and automated surveillance to operate seamlessly in real time.

With localization techniques such as Wi-Fi-based positioning systems (WPS) and the advent of convolutional neural networks (CNNs), we can detect and extract spatial patterns from visual data in a remarkable manner. This has the potential to transform industries, including healthcare, retail, automotive, and traffic management. Combining the high-processing capabilities of cloud computing with the real-time features of edge computing and the connectivity enhancements brought about by Wi-Fi technologies can enable AI to lift computer vision to a different level. Consider industrial automation in large factories and warehouses, smart cities with optimized traffic control, retail centers and shopping complexes with enhanced services and store layouts based on monitored customer behavior, and even remote medical

services and patient monitoring that enable accurate diagnosis in real time. All these developments can take place by leveraging computer vision that is powered by AI at the edge.

While there are still challenges standing before computer vision today, such as high processing power and high-quality data, the potential that Edge AI brings, supported by Wi-Fi capabilities that ensure fast and seamless data transfer - even in remote areas where accessibility would otherwise be difficult - will propel computer vision and enable it to be a much more potent and ubiquitous technology. With on-device inference capabilities and functionalities that enable computer vision within edge and mobile devices, image and pattern recognition can improve significantly,

taking advantage of edge computing's reduced latency, enhanced security, and minimized costs. Integrating AI into the mix enables even low-cost cameras to act as smart cameras, with deep learning methods enabling better identification and interpretation of data patterns. This is further enhanced by leveraging data from multiple devices connected via Wi-Fi, which provides AI with more comprehensive data, reinforcing its precision and understanding.



The impact of computer vision developments on AIoT. Credit: Softweb Solutions

Conclusion

Wi-Fi-Powered Edge AI - A Technology for the Future

Wi-Fi and Edge AI represent a technological paradigm shift with far-reaching implications. The convergence of Wi-Fi with Edge AI is not just a technological evolution but a fundamental reshaping of the digital landscape. As this symbiotic relationship matures, it propels us towards a future where Edge AI is not just a technology but an integral and indispensable part of our interconnected world. The potential for transformative applications is immense, urging both developers and sellers to explore the uncharted territories of innovation and contribute to the unfolding narrative of Edge AI's promising future.

This not only addresses the computational needs of Edge AI but also aligns with the growing trend of connected devices in smart buildings, industrial settings, healthcare, retail, and smart cities. The integration of Edge AI with Wi-Fi services brings forth a boost in performance, user experience, and security measures, ensuring that sensitive data is

processed locally. This transformation leads to the development of robust, always-on systems capable of meeting the stringent privacy requirements of contemporary applications and the ever-growing need for better performance, flexibility, scalability, and reliability.

Throughout this report, we explored the capabilities and impact of Edge AI, especially when coupled with Wi-Fi services. We highlighted the valuable insights provided by Relay2, the leading Wi-Fi technology provider for the Edge AI market. Relay2's Service Points technology is designed to deliver exceptional application performance while enhancing security, privacy, and scalability at the edge. The Relay2 ServiceEdge Platform harnesses the power of Wi-Fi Services with Edge Computing to create a digital world where application performance and user experience not only remain uncompromised but are also enhanced. We also dove into particular applications and reviewed some use cases of Wi-Fi services edge computing. And finally, we took a look at the future of these technologies and how they will impact our businesses and way of living in the coming years. All this could not have been done without the support of Relay2 in developing this report.

Relay2 is committed to empowering its partners by providing comprehensive, innovative solutions to ensure mutual success leveraging Wi-Fi Services with Edge Computing and the Relay2 ServiceEdge Platform. Therefore, developers and sellers are encouraged to partner with Relay2 to unlock the full potential of technology and shape a seamless digital future. The convergence of Wi-Fi with Edge computing can reshape the digital technology and business landscape, and Relay2 is at the forefront of this transformation.

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References

1

Keen, E. (2023). Gartner Identifies Top Trends Shaping the Future of Data Science and Machine Learning. *Gartner*. <https://www.gartner.com/en/newsroom/press-releases/2023-08-01-gartner-identifies-top-trends-shaping-future-of-data-science-and-machine-learning>

2

PR Newswire (2023). Edge AI Market to Hit \$66.47 Billion by 2030 at CAGR of 21.0%: Grand View Research, Inc. *Yahoo Finance*. <https://finance.yahoo.com/news/edge-ai-market-hit-66-093000634.html>

3

Gaikwad, V. and Rake, R. (2021). AI Edge Computing Market. *Allied Market Research*. <https://www.alliedmarketresearch.com/ai-edge-computing-market-A14885>

4

Wadhvani, P. (2023). Edge AI Market. *Global Market Insights*. <https://www.gminsights.com/industry-analysis/edge-ai-market>

5

Fortune Business Insights (2022). Edge AI Market. <https://www.fortunebusinessinsights.com/edge-ai-market-107023>

6

Perri, L. (2023). What's New in Artificial Intelligence from the 2023 Gartner Hype Cycle. *Gartner*. <https://www.gartner.com/en/articles/what-s-new-in-artificial-intelligence-from-the-2023-gartner-hype-cycle>

7

van der Meulen, R. (2018). What Edge Computing Means for Infrastructure and Operations Leaders. *Gartner*. <https://www.gartner.com/smarterwithgartner/what-edge-computing-means-for-infrastructure-and-operations-leaders>

8

Haque, A. et al. (2020). Illuminating the dark spaces of healthcare with ambient intelligence. *Nature*. 585, 193-202.

9

Nguyen Gia, T. et al. (2019). Edge AI in Smart Farming IoT: CNNs at the Edge and Fog Computing with LoRa. *IEEE AFRICON*. 1-6. doi: 10.1109/AFRICON46755.2019.9134049.

10

MIT Sloan ME Staff (2024). Edge AI Will Start to Transform Business in 2024. *MIT Sloan ME*. <https://www.mitsloanme.com/article/edge-ai-will-start-to-transform-business-in-2024/>

About Relay2

Relay2 is at the forefront of Wi-Fi Services with Edge Computing, driven by our revolutionary ServiceEdge Platform. Our breakthrough device technology, known as Service Points, is designed to deliver exceptional application performance while also enhancing security, privacy, and scalability at the edge. We harness the power of Wi-Fi Services with Edge Computing to create a digital world where application performance and user experience remain uncompromised. Together, we can unlock the full potential of technology and shape a seamless digital future. Relay2 is committed to empowering partners by providing comprehensive, innovative solutions to ensure mutual success leveraging Wi-Fi Services with Edge Computing and the Relay2 ServiceEdge Platform.



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